

NITIN KUMAR

QA & Manufacturing Engineer | AI Automation · Data · IoT · Robotics

Gurugram, Haryana, India | +91 75368 55614 | ernitinkumar14@gmail.com

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Portfolio

PROFESSIONAL SUMMARY

Quality engineer with **9+ years** in medical-device and precision manufacturing and currently pursuing an **M.Tech in Applied Mechatronics & Robotics (IIT Bhilai, 2025 – 2027)**. Experienced in ISO 13485 quality systems, validation (IQ/OQ/PQ), supplier quality, CAPA and audits. Builds practical Python, data and IoT tools that remove repetitive work and make engineering data easier to act on. GATE 2024 qualified (XE).

KEY HIGHLIGHTS

- **100%** ISO 13485 compliance maintained across **4** production lines; **zero** major audit findings in 4 years
- **15%** production-efficiency gain through validation of **12+** new product launches
- Managed **30+** suppliers at **95%** on-time delivery with zero major non-conformances
- Designed an automated leak / pressure-decay test device, eliminating manual testing errors

PROFESSIONAL EXPERIENCE

IPQC Engineer / Quality Inspector

Feb 2025 – Present

SS Innovations Pvt. Ltd., Gurugram, Haryana

- In-process quality control, incoming inspection and first-piece inspection in medical-device manufacturing
- Inspection and evaluation of robotics-related components, including robot arm parts and Tool Interface (TI)
- Measurement with Vernier caliper, micrometer and bore gauge; plating thickness inspection
- Quality documentation, process quality monitoring, root cause analysis and CAPA-related activities

Senior Quality Engineer

Jun 2020 – Feb 2025

Allied Medical Limited, Bhiwadi, Rajasthan

- Managed QA for OT equipment: surgical tables, operating lights, patient monitors and precision devices
- Maintained 100% compliance with ISO 13485 and international regulatory standards across 4 production lines
- Boosted production efficiency by 15% via process validation and verification for 12+ new product launches
- Reduced manufacturing error rates through cross-functional quality risk mitigation programs
- Designed a mechanical assembly testing device for leak and pressure testing (pressure decay method)
- Managed supplier quality for 30+ vendors: 95% on-time delivery, zero major non-conformances
- Wrote IQ/OQ/PQ protocols, implemented SPC, and led internal and external audits with zero major findings
- Maintained DHR, DMR, batch records and validation reports to ISO 13485, 21 CFR Part 820, BIS and CDSCO standards

QC Inspector

Jul 2017 – May 2020

Allied Health Technology Ltd, Gurugram, Haryana

- Validated dimensional accuracy of 5,000+ CNC/VMC/lathe-machined medical components with a 100% approval rate
- Improved product yields by 10% through rejection analysis and corrective actions; reduced scrap by 8%
- Held tolerances of ± 0.01 mm; performed FAI, final inspection, NCR/CAPA, GD&T interpretation and AQL sampling

EDUCATION

M.Tech, Applied Mechatronics & Robotics

2025 – 2027

Indian Institute of Technology (IIT) Bhilai

Mechatronics, robotics, automation, embedded systems, IoT, control systems

B.Tech, Mechanical Engineering

2013 – 2017

Dr. A.P.J. Abdul Kalam Technical University (AKTU), Uttar Pradesh

GATE 2024: Qualified

Engineering Sciences (XE)

TECHNICAL SKILLS

Quality & Regulatory: ISO 13485, ISO 9001:2015, ISO 14971, 21 CFR Part 820, CDSCO, BIS, GMP, IQ/OQ/PQ, Process Validation, Supplier Quality, CAPA, RCA (5-Why, Fishbone), FMEA, SPC, MSA, PPAP, GD&T, CMM Metrology, DHF/DMR/DHR, Change Control, Internal & External Audits, Lean Six Sigma, OEE & Quality Dashboards

AI & Data: Python, Pandas, NumPy, SQL, Streamlit, Data Visualization, Excel & Report Automation, Generative AI, Prompt Engineering, AI API Integration; learning Machine Learning (Scikit-learn)

IoT & Embedded: Raspberry Pi, Arduino, ESP32, NodeMCU, sensor integration, I2C, UART, IoT monitoring

Robotics & Electronics: ROS2, Mechatronics, Control Systems, Actuators, Microcontrollers, Basic Electronics, MOSFET, SMPS, DC-DC conversion

SELECTED PROJECTS

- **Smart OT Environmental Monitoring System** (in development): Raspberry Pi with SCD30, BME280 and PM sensors, data logging, dashboard and threshold alerts for operating-theatre conditions
- **AI Quality Data Analyzer:** Python/Streamlit workflow turning inspection data into rejection trends, Pareto charts and an AI-drafted management summary
- **AI Report Automation:** Excel reporting automated with Python, charts, AI summary and PDF/HTML export
- **IoT Environmental Monitoring:** Raspberry Pi sensor logging, dashboard and alerts
- **Arduino Day/Night & Rain Detection; ROS2 Robotics** (nodes, topics, services, actions, launch files)

CERTIFICATIONS

- Entrepreneurship Development Programme (EDP), Institute For Industrial Development (Samadhan, Ministry of MSME, Govt. of India), Jul 2026
- IoT and Robotics, Launched Global (Jul 2026)
- Lean Six Sigma Green Belt (Apr 2025) and Yellow Belt (Jul 2024), The Knowledge Academy
- Python Workshop, United Latino Students Association (Sep 2025); Arduino Programming, Udemy (Jul 2025); Advanced Excel, Udemy (Dec 2022)
- AutoCAD + CNC Milling, Ministry of MSME, Government of India (2015)